

# Qoneqt Video Factory

*Topic in. Global Feed short out.*

Theme: Build an LLM-Powered Content Pipeline for Qoneqt. A repeatable system that turns a topic into a publish-ready, captioned vertical video for the Qoneqt Global Feed.

TEAM

**ProCoders**

COLLEGE

LJ University

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Three shorts generated by the pipeline tonight:  
Tech, Motivation, Hinglish presets

# The Global Feed is hungry. Video is slow.

## What is the problem?

A social feed lives on fresh video every day. Making one 30-second short by hand means writing a script, finding visuals, recording a voice, editing, and adding captions. A creator manages one a day. The Global Feed needs hundreds.

**2-4 hrs**

to hand-make one 30-second short

**0**

Qoneqt-native topic-to-video tools

Result: thin, repetitive video content, creators who burn out, and communities that go quiet between posts.

?

### Who is affected

Qoneqt creators and community admins who must post daily, and the platform, whose Global Feed runs on steady video.

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### Current solutions

Manual editing apps (CapCut, Canva), one-shot AI video generators on paid credits, or hiring an agency. All produce one video at a time.

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### Key gaps

- Not repeatable: no batch, no pipeline
- Not community-aware: generic tone
- Paid per video; no enforced hook or captions

# A video factory, not a video button

Type a topic. Pick a community. Get a publish-ready short and its post text in about a minute.

## 1 Plan

The LLM writes 3 hooks, scores them, keeps the best, then a 5–7 scene script with a stock-search query per scene, plus caption and hashtags.

## 2 Produce

An AI voice reads each scene. An AI still is generated per scene, or a photo fetched, then given motion. Whisper gives word timings for word-pop captions.

## 3 Publish

ffmpeg assembles a 1080×1920 MP4 with burned-in captions. The app hands you the video, caption, and hashtags ready to paste into Qoneqt.

## What makes it better than existing options

|                           | Manual editing | One-shot AI tools | Video Factory                         |
|---------------------------|----------------|-------------------|---------------------------------------|
| Repeatable at scale       | No             | No                | <b>Yes, batch of 10 per click</b>     |
| Community-aware tone      | By hand        | No                | <b>6 presets, one dict entry each</b> |
| Hook, captions, post text | By hand        | Partly            | <b>Built in, every video</b>          |
| Cost per video            | Hours          | Paid credits      | <b>₹0 on free tiers</b>               |

# Input → Process → Output



Every stage has a fallback chain, so one flaky free API never kills a job. A single worker thread drains the job queue; batch mode is just N queued topics.

## Frontend

One static HTML page. Textarea for topics (one per line), community dropdown, live stage progress, inline player, Copy post text, Download.

## Backend

FastAPI. POST /generate, GET /jobs, static /out. In-memory job dict, one worker thread, files per job in out/<id>/.

## AI layer

LLM for plan JSON (validated, guided retry). TTS per scene. Photo or clip search per scene. Whisper word timings aligned to script words.

## Data + deploy

Outputs as files (mp4, jpg, json). Docker image with ffmpeg and Noto fonts. Deployed on Railway, keys as secrets.

**Stack** Python 3.11 · FastAPI · httpx · ffmpeg · Docker · Groq (gpt-oss-120b, Whisper large-v3-turbo) · Pollinations FLUX · Railway · ElevenLabs · edge-tts · Wikimedia Commons

# Five things that make it work

## 1 Community presets

Six Qoneqt communities (General, Tech, Fitness, Motivation, Finance, Hinglish Fun). Each sets tone, language, narrator voice, caption style, hashtags, and caption colour. Adding one is a single dict entry.

## 2 Self-scored hooks

The LLM writes three opening lines, scores each for scroll-stopping power, and the best one opens the video. Quality control inside the prompt, zero extra calls.

## 3 Word-pop captions

Whisper word timestamps drive karaoke-style captions: each spoken word lights up in the preset colour, aligned back to the script's own spelling. The look that performs on shorts.

## 4 Batch and fallbacks

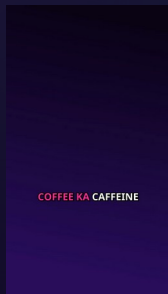
Ten topics per click, one worker, no babysitting. Every stage has a fallback (LLM, voice, stock, captions), so the factory keeps producing when a free API blinks.

## 5 Post-ready output

Not just an MP4. Hook, caption, and hashtags come out together, with a one-click Copy post text button, so publishing to Qoneqt is a 30-second manual step.

## Real output frame

Hinglish preset, edge-tts fallback voice, Whisper-timed captions, pink accent word. Generated, not mocked.



# Impact and feasibility

₹0

API cost per video on free tiers

~1 min

to make a 30–45 s short  
(measured tonight)

10

topics per click in batch mode

6

community presets, one dict entry to add more

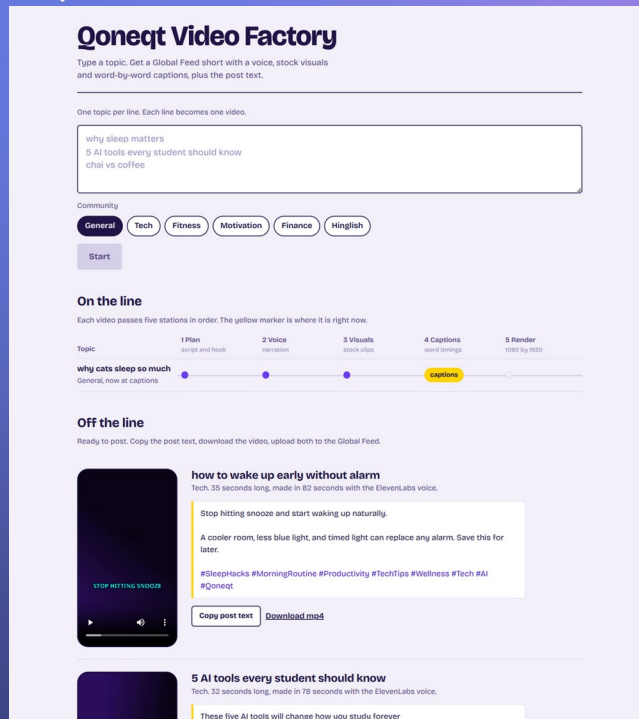
## Who benefits and how

- Creators: turn ideas into daily shorts without editing skills or hours of work
- Community admins: on-tone videos in their own voice, Hinglish included
- Qoneqt: a steadier, higher-quality stream on the Global Feed, and a tool it can offer creators natively
- Real-world: 100 topics in overnight, 100 publish-ready videos out

## Scalability and feasibility

- One free CPU: ~40–60 videos/hour per worker; scale by adding workers
- Free tiers: Groq LLM + Whisper, ElevenLabs (edge-tts fallback), Wikimedia visuals
- One Docker container on Railway; keys stay secret
- Working end to end today and live on a public URL

# Built and running tonight



## What the prototype does today

- Batch of topics queued from the page; each runs plan → images → voice → visuals → captions → render with live progress
- 34-second videos in about 55 seconds each, word-pop captions, post text ready to copy

## Try the live demo in 3 steps

- 1 **Write a prompt** e.g. "AI use cases in daily life" (one topic per line)
- 2 **Select a category** pick a community: Tech, Fitness, Hinglish Fun...
- 3 **Press Start** watch the stages run; the video is ready in ~1 min



**Open Live Demo**

[qoneqt-video-factory.up.railway.app](https://qoneqt-video-factory.up.railway.app)

*Qoneqt doesn't need one AI video. It needs a factory. We built the factory.*